Data Science

Regression Metrics

Machine Learning Fundamentals

GENERAL ASSEMBLY

Lesson roadmap

Welcome + Introduction	What Are Metrics?	Loss Functions	Common Metrics	Using Metrics to Assess Model Performance	Bring It Home

Learning objectives

1. Describe the principles of Regression Metrics.
2. Define common metrics including error measures (MAE, MSE, RMSE), R2, Adj R2 and others.
3. Apply regression metrics to assess OLS models.

What Are Metrics?

Comparing Models

You build two OLS models to predict sales.

➔ **Model 1**: includes 3 independent variables on TV ad spend, Radio ad spend, and Social ad spend.

➔ **Model 2**: includes 3 independent variables on Facebook ad spend, TV ad spend, and LinkedIn ad spend.

How do we assess which model is performing better?

Comparing Models

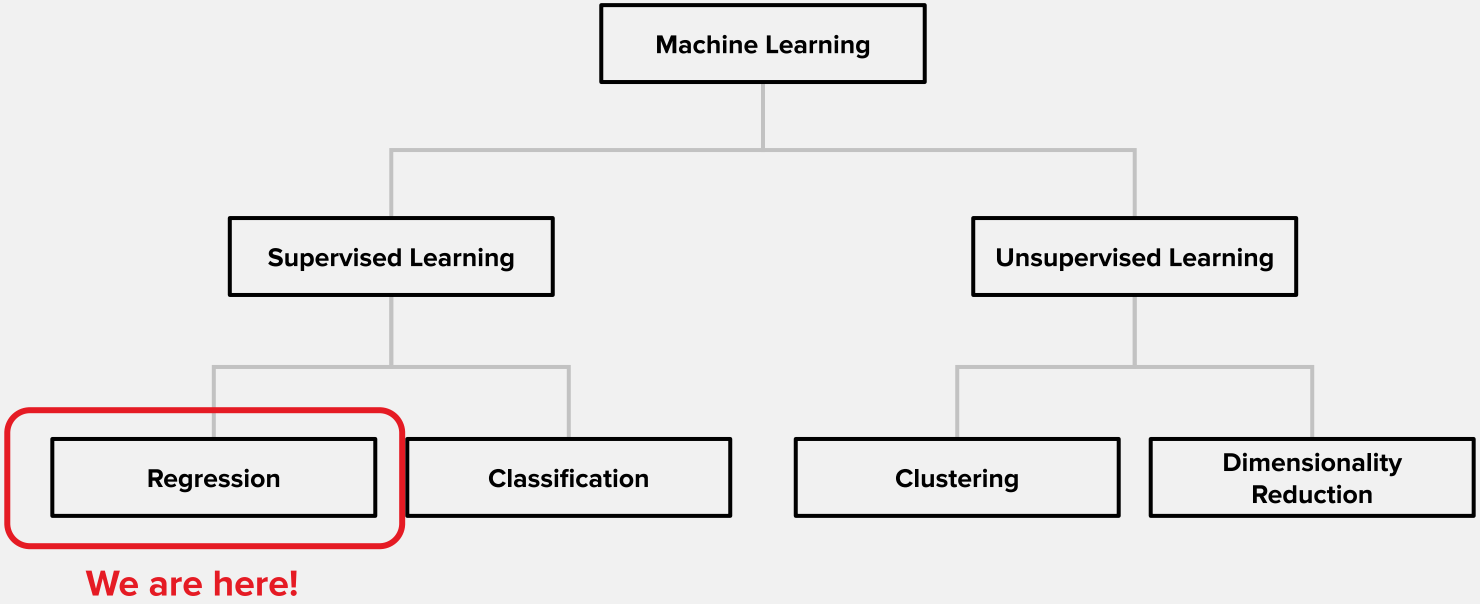
How do we assess which model is performing better?

Metrics!

Different types of models require different metrics

- Regression and classification models use different metrics.
- There are also specialized metrics for certain models and tasks (such as Translation NLP models).

Where are we now?



What is a metric?

1

A metric is short for "distance metric".

2

Provides a numerical measure of distance between two things.

3

Has a strict mathematical definition that enables comparison across models and use cases

Four Rules of Metrics

For any two vectors x and y, a metric must satisfy:

1. Distances must be positive: `d(x,y) ≥ 0`
2. 0 distance means the two items are identical: `d(x,y) = 0` if and only if `x = y`
3. Distances are symmetric: `d(x,y) = d(y,x)`
4. Triangle Inequality: `d(x,y) ≤ d(x,z) + d(z,y)`

Why does this matter?

Goal: Get our predictions ($\hat{y}$) as close as possible to the truth (y).

Predictions come from the models we create while the “truth” is the labeled values we use to train our models.

Loss Functions

Recall our goal?

Goal: Get our predictions ($\hat{y}$) as close as possible to truth (y).

- We need to **minimize the distance between them**.
- **Metrics** give us a way to **measure this distance**.

Predictions come from the models we create while the “truth” is the labeled values we use to train our models.

Loss Function Terms

A **loss function** (aka, a **distance metric**) computes a **loss metric** or **error metric**.

The process of computing a loss function is sometimes called **scoring**. If the loss function was also used to fit the model, then it's also the model's **objective function**.

But don't worry about this… everyone just kinda uses all of these interchangeably in practice.

Loss Function

A **loss function** is a **distance metric** that describes how far our predictions ($\hat{y}$) are from the ground truth (y). We seek to minimize these loss metrics.

Most ML algorithms are actually fit by finding the best set of parameters that minimize a specific loss metric.

Common Metrics

Trivia

What is the loss metric used for OLS?

- **A** Residuals
- **B** Mean Squared Error
- **C** Sums
- **D** Least Regressions

The Sum Squared Error

One aggregate measurement of the quality of our OLS model is the **sum squared error**, or **SSE**:

$$SSE = \sum (y_i - \hat{y}_i)^2 = \sum e_i^2$$

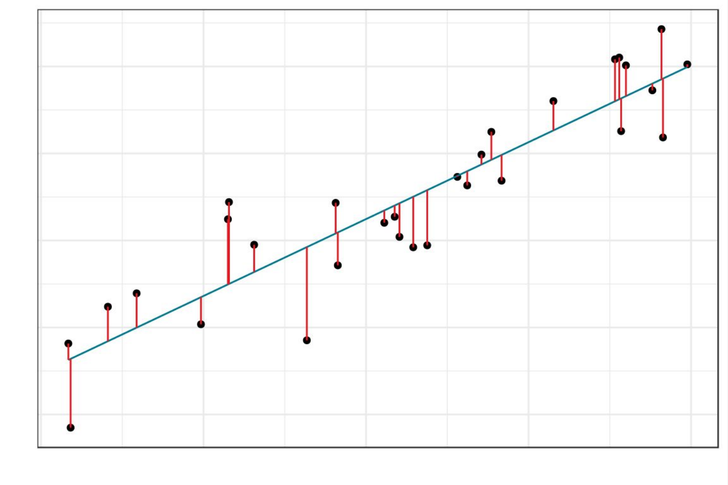
Or more commonly, the **mean squared error (MSE)**:

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum e_i^2$$

Fitting OLS models

Remember, this is the quantity we actually seek to **minimize** in order to find the best values of our betas!

$$MSE = \frac{1}{n}\sum(y_i - \hat{y}_i)^2 = \frac{1}{n}\sum(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$



The Sum Squared Error

ADVANTAGES	DISADVANTAGES
→ It's literally what OLS uses to fit the best model anyway! → It's foundational for computing other error metrics below. → You can still compare SSEs between models (lower is better). → Punishes "big" residuals more harshly. → DIFFERENTIABLE! < Big deal later!	→ Not human interpretable. → Punishes "big" residuals more harshly. → Units are square. → Sensitive to outliers.

The Mean Squared Error

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→ It's foundational for computing other error metrics below.	→ Units are square.
→ You can still compare SSEs between models (lower is better).	→ Sensitive to outliers.
→ Punishes "big" residuals more harshly.	
→ DIFFERENTIABLE! < Big deal later!	~~→ Not human interpretable.~~
→ **Slightly more human interpretable**.	

Root Mean Square Error

If the MSE is sort of like a variance, we might like to also look at its square root, which would then be sort of like a standard deviation:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)\right)^2}$$

Root Mean Square Error

ADVANTAGES	DISADVANTAGES
→ Similar to MSE Advantages. → Widely used. → More interpretable due to similarity to standard deviation.	→ Punishes "big" residuals more harshly. → Sensitive to outliers.

R^2 Higher Values: Closer to 1 is generally better

Another popular regression metric is the R2, or the **coefficient of determination**.

It is the proportion of variability in **y** we can explain with **x**, in other words the proportion of variation your model explains.

Possible values vary from 0 to 1: if R^2 is 1, that means all of the variation is accounted for by the model. **Higher values (closer to 1 is generally better)**.

$$R^2 = 1 - \frac{\text{SSE}}{\text{Null SSE}} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Source

Adjusted R^2

R^2 will increase as you add features to your model. So a model with 100 features will have a higher R^2 than a model with 10.

Adjusted R^2 attempts to account for this and is the recommended metric when comparing regression models with different numbers of features. Similar to R^2.

Higher values (closer to 1 is generally better).

Source

Other Metrics

There are also a ton more metrics that we won't discuss.

These are only valid for OLS, and their goal is to penalize the metric if too many x-variables are used.

We have better tools for this now, so these are considered a bit old-fashioned. They're not even included with **scikit-learn**!

Source

Examples of Regression Metrics

1. Mallows's C□
2. Akaike Information Criterion (AIC)
3. Akaike Information Criterion correction (AICc)
4. Bayesian Information Criterion (BIC)

Using Metrics to Assess Model Performance

Compare different versions of models in `scikit-learn`

Use the metrics on the next slide to compare the performance of various models and select the one to use for your use case.

Source

Compare different versions of models in `scikit-learn`

ERROR-BASED METRICS	EXPLAINED VARIANCE METRICS
Mean Squared Error (MSE): `metrics.mean_squared_error` ○ Sensitive to outliers, penalizes larger errors heavily ○ Lower is better; range: [0, ∞) **Root Mean Squared Error (RMSE)**: `np.sqrt(metrics.mean_squared_error)` ○ Same units as target variable, more interpretable than MSE ○ Lower is better; range: [0, ∞) **Mean Absolute Error (MAE)**: `metrics.mean_absolute_error` ○ Less sensitive to outliers than MSE/RMSE ○ Lower is better; range: [0, ∞)	**R^2 (Coefficient of Determination)**: `metrics.r2_score` ○ Proportion of variance explained by the model ○ Higher is better; range: (-∞, 1] ○ Can be negative if model is worse than mean prediction

When should you use each metric?

Generally, you would want to use **one error metric** (MSE, RMSE, MAE) and **one variance metric** (such as R2 or adj R2) when you are doing any regression style model.

Which specific metric should you use?

it doesn't have a major impact which **error metric** you choose. Generally, the one you use is determined by your field/sector. Use the same metric as your colleagues, and use the same metric throughout your modeling build process.

For **variance metrics**, if you are comparing models with different numbers of features use adj R2. Otherwise R2 is great.

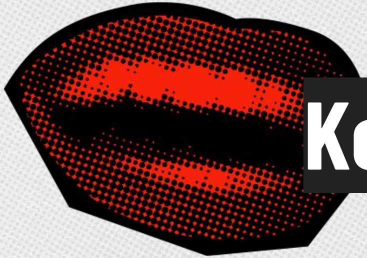
Source

Regression Metrics

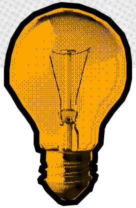
Let's see it for ourselves!

- Open **02-regression-metrics-starter-code.ipynb**

Bring It Home



Key takeaways

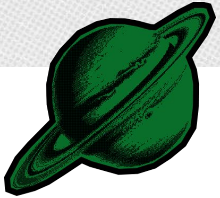


Metrics assess model performance

There are many metrics for specialized use cases.

Regression and classification metrics are different

Use the correct metrics for your model.



No metric is perfect

Beware of overfitting and bias which aren’t easily assessed with metrics.

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